

Fredrick George F

Backend Software Engineer | Data Science Undergraduate

+91 8792938911 | fredrick.george.f.25@gmail.com | Bengaluru, India

LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in Backend Architecture and Data Science. Experienced in engineering strictly layered, ACID-compliant systems prioritizing clean separation of concerns, and maintainability. Focused on structured software engineering, manual database lifecycle management, and integrating Generative AI into highly scalable full-stack applications.

EDUCATION

New Horizon College of Engineering

B.E. in Computer Science (Data Science) | CGPA: 8.62 / 10.0

Bengaluru, India

2023 – 2027

St. Francis Composite PU College

Pre-University Course (PUC) | Percentage: 88.5%

Bengaluru, India

2022 – 2023

TECHNICAL SKILLS

Languages: Java (Advanced), Python (Intermediate), SQL (Advanced), JavaScript (Intermediate), HTML, CSS

Tools & Frameworks: Spring Boot, Django, Next.js, React.js, JDBC, PostgreSQL, Redis, JWT, Git, Tailwind CSS

SOFT SKILLS

Technical Communication, Cross-Functional Collaboration, Team Mentorship

EXPERIENCE & LEADERSHIP

AICTE IDEA Lab, New Horizon College of Engineering

Bengaluru, India

Technical Intern & Volunteer Coordinator

May 2025

- Operated advanced lab machinery and engineered hands-on STEM projects to facilitate a Summer Camp for 50+ participants.
- Mentored students in Web technologies, coordinating with AICTE officials to ensure seamless technical resource allocation and a 100% active engagement rate.

CORE ENGINEERING PROJECTS

Authentication Microservice (auth_system) | Java, Spring Boot, Redis, JWT

Mar 2026 – Present

- Designed a stateless authentication service using HS256-signed JWTs, leveraging Spring Security filter chains to enforce fine-grained, path-based RBAC (/admin, /user), eliminating server-side session dependency.
- Implemented a secure cross-device QR login flow using ephemeral Redis sessions (120s TTL) with atomic state transitions, preventing replay attacks and ensuring single-use authentication.
- Optimized token validation and authorization pipelines for low-latency request processing under concurrent access.

Student Management CLI System and API | Java, JDBC, PostgreSQL

Feb 2026

- Designed a layered backend architecture using JDBC (without ORM) to retain fine-grained control over SQL execution, transaction boundaries, and performance.
- Implemented atomic CRUD operations with explicit transaction management (commit/rollback), ensuring consistency and failure-safe execution.
- Built an immutable audit logging system within transaction scopes to guarantee complete traceability of all database mutations.

Farmers' Market Connect (Agricultural Marketplace MVP) | Django, React, PostgreSQL

Jun 2025

- Led backend development of a multi-module Django REST API platform connecting 100+ farmers with retailers for direct marketplace interaction.
- Implemented JWT-based authentication and role-based access control to secure multi-user workflows.
- Developed a responsive React SPA with protected routing and modular UI architecture for seamless user experience.

MINI PROJECTS

Unisphere LMS (Educational Platform) | React.js, TypeScript, Supabase

May 2025

- Built a role-based LMS interface with Supabase authentication and Google Classroom OAuth2 integration.

PlacementPulse | React.js, TypeScript, Supabase

Jan 2026

- Used a streak-based, gamified system for consistent DSA and aptitude practice, strengthening problem-solving discipline and improving structured approach to technical questions.

ArNav (Voice-Guided Navigation for the Blind) | Android, Java

Apr 2026

- Developed a voice-guided navigation prototype using Speech-to-Text and Text-to-Speech APIs and Google Map API and leveraging Gemini API for natural speech navigation and employed haptic feedback for secondary output.

CERTIFICATIONS & ACHIEVEMENTS

- Innovation Ambassador, MoE's Innovation Cell** (Mar 2026) – 30 hours of formal training in design thinking and enterprise innovation.
- Generative AI Landscape, Infosys Springboard** (Mar 2026) – Verified completion on LLMs and enterprise AI applications.
- Python Essentials 1, Cisco** (Nov 2025) – Completed formal training with distinction (Score: 90.00%).
- AICTE IDEA Lab R&D Skill Program** (Oct 2025) – Advanced technical training CNC laser and Additive manufacturing technologies.